

Mensuration

2- D: Under 2 dimensions we will study

- Triangles
- Quadrilateral
- Polygons
- Circle

3- D: Under 3 dimensions we will study

- Cube
- Cuboids
- Box
- Cylinder
- Prism
- Cone
- Pyramid
- Frustum
- Sphere
- Hemisphere
- Tetrahedral

Note: For 2D figures, area and perimeter are calculated and for 3D figures, volume and surface area are calculated.

TWO DIMENSIONAL FIGURES

TRIANGLE

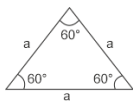
For a triangle with height, h, and base, b;

Area = $\frac{1}{2} \times b \times h$ and Perimeter

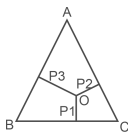
= Sum of all sides

Special cases:

1) Equilateral Triangle - All sides are equal and each angle is 60° .



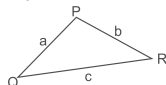
- Area = $\frac{\sqrt{3}}{4} a^2$
- Height = $\frac{\sqrt{3}}{2} a$
- Perimeter = $3a$
- Inradius(r) = $\frac{a}{2\sqrt{3}}$
- Circumradius(R) = $\frac{a}{\sqrt{3}}$



If P1, P2 and P3 are perpendicular to a side from a point inside the triangle, then

- Height of triangle = P1 + P2 + P3

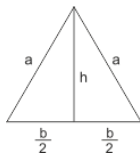
2) Scalene Triangle - Sides are of unequal lengths.



Area = $\sqrt{s(s-a)(s-b)(s-c)}$, This formula is called Heron's formula.
Where, Semi-perimeter,
 $s = \frac{a+b+c}{2}$

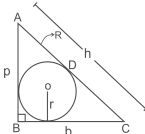
- Perimeter = $a + b + c$

3) Isosceles Triangle :- two sides and two angles are equal. Altitude bisects the base.



- Area = $\frac{b}{4} \sqrt{4a^2 - b^2}$
- Height = $\sqrt{a^2 - \left(\frac{b}{2}\right)^2}$
 $= \frac{1}{2} \sqrt{4a^2 - b^2}$
- Perimeter = $a + a + b$
 $= 2a + b$

4) Right angled Triangle :- One of the angles is 90° .



Here, p = perpendicular, b = base and h = hypotenuse

- Area = $\frac{1}{2} \times b \times p$
- Perimeter = $p + b + h$
- Pythagoras Theorem: $h^2 = p^2 + b^2$
- Inradius(r) = $\frac{p+b-h}{2}$
- Circumradius(R) = $\frac{h}{2}$

Note: Common Pythagoras triplets:

(1, 1, $\sqrt{2}$); (1, 2, $\sqrt{5}$); (3, 4, 5); (5, 12, 13); (7, 24, 25); (20, 21, 29); (9, 40, 41); (8, 15, 17) (12, 35, 37) (60, 11, 61) (65, 72, 97) (96, 110, 146).

Try remembering them.

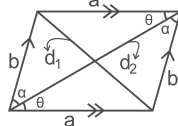
QUADRILATERAL

A figure enclosed by four sides is called a quadrilateral. A quadrilateral has four

angles and sum of these angles is equal to 360° .

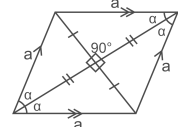
Special Cases:

1) Parallelogram - It is a quadrilateral with opposite sides parallel and equal.



- Area = base $\times$ height
- Perimeter = $2(a + b)$
- $d_1^2 + d_2^2 = 2(a^2 + b^2)$
- In a parallelogram, opposite sides are equal, opposite angles are equal and diagonals bisect each other.

2) Rhombus - It is a parallelogram with all four sides equal. The opposite angles in a rhombus are equal.

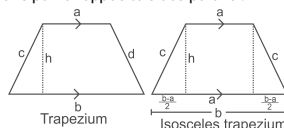


Here, a = side; d_1 and d_2 are diagonals.

- Area = $\frac{1}{2} \times d_1 \times d_2$
- Side (a) = $\frac{1}{2} \sqrt{d_1^2 + d_2^2}$
- Perimeter = $4a$
- $4a^2 = d_1^2 + d_2^2$

Diagonals bisect each other at right angles.

3) Trapezium - It is a quadrilateral with one pair of opposite sides parallel.



Here, a and b are parallel sides and h is the height or perpendicular distance between a and b.

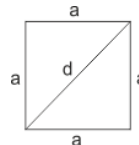
- Area = $\frac{1}{2} \times \text{height} \times (\text{sum of parallel sides})$
 $= \frac{1}{2} \times h \times (a + b)$

4) Rectangle - It is a parallelogram with equal opposite sides and each angle is 90° .

- Area = Length $\times$ Breadth = $L \times B$

- Perimeter = $2(L + B)$
- Diagonal (d) = $\sqrt{L^2 + B^2}$
- Area of the walls of a rectangular room = $2 \times (L + B) \times H$

5) Square - It is a parallelogram with all four sides equal and each angle is equal to 90° .



- Area = $(\text{side})^2 = a^2 = \frac{1}{2} d^2$
- Perimeter = $4 \times \text{side} = 4a$
- Diagonal (d) = $a\sqrt{2}$

NOTE: Important points about Quadrilaterals

1. The diagonals of a parallelogram bisect each other.
2. Diagonal of a parallelogram divides it into two triangles of equal area.
3. The diagonals of a rectangle are of equal lengths and bisect each other.
4. The diagonals of a square are equal and bisect each other at right angles.
5. A rhombus has unequal diagonals and they bisect each other at right angles.
6. A parallelogram and a rectangle have equal areas if they are on the same base and between the same parallel lines.

REGULAR POLYGON

In a regular polygon all sides and all interior angles are equal. A polygon is called a pentagon, hexagon, heptagon, octagon, nonagon and a decagon as they have 5, 6, 7, 8, 9, 10 sides, respectively. If each side of a regular polygon of 'n' sides is equal to 'a' then:

- Area of regular pentagon = $5 \times \frac{\sqrt{3}}{4} a^2$
- Area of regular hexagon = $6 \times \frac{\sqrt{3}}{4} a^2$
- Each exterior angle = $\frac{360^\circ}{n}$
- Each interior angle = $180^\circ - \text{Exterior angle}$
- Number of diagonals = $\frac{n(n-3)}{2}$
- Sum of all interior angles = $(n-2) \times 180$
- Sum of all exterior angle = 360°

CIRCLE

It is a plane figure enclosed by a line on which every point is equally distant from

a fixed point (centre) inside the circle.



- Area = πr^2
- Circumference (perimeter) = $2\pi r$
- Diameter = $2r$
- Length of Arc (AB) = $2\pi r \times \frac{\theta}{360^\circ}$
- Area of sector AOB = $\pi r^2 \times \frac{\theta}{360^\circ}$
- Length of Arc (AB) = $\text{angle} \times \text{radius}$, (where angle in radian)

CIRCULAR RING



Here, R = radius of bigger ring, r = radius of smaller ring

- Area = $\pi(R^2 - r^2)$
- Difference in circumference of both the rings = $2\pi R - 2\pi r$

Short Cut methods/Tricks

1. If the length and breadth of a rectangle are increased by a% and b%, the area of the rectangle will be increased by $(a + b + \frac{ab}{100})\%$
2. If any of the two sides of the rectangle is decreased then use '-ve' values for a.
3. All the sides of any two dimensional figure changed by a%, then its area will

change by $(2a + \frac{a^2}{100})\%$

Whenever there is a decrease, use negative value for 'a'

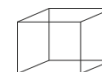
4. If all the sides of any two dimensional figure has changed (increased or decreased) by a% then its perimeter also changes by a%. In the case of a circle such changes take place because of the change in radius (or diameter).
5. If the area of a square is 'a' square unit. Then the area of the circle formed with the same perimeter is given by square units.
6. Area of a square inscribed in a circle of radius 'r' is equal to $2r^2$.
7. The area of the largest triangle inscribed in a semicircle of radius r is equal to r^2 .
8. If a pathway of width x is made inside or outside a rectangular plot of length L and breadth B, then area of the pathway is

- (i) $2x(L + B + 2x)$, if path is made outside the plot
- (ii) $2x(L + B - 2x)$, if a path is made inside the plot.

9. If two paths, each of width x are made parallel to length (L) and breadth (B) of the rectangular plot in the middle of the plot crossing each other, then, Area of the path = $x(L + B - x)$

THREE DIMENSIONAL FIGURES

CUBE



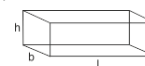
All sides are equal. It has six faces and 12 edges.

- Volume = a^3
- Total surface area = $6a^2$
- Diagonal = $a\sqrt{3}$
- Sum of all edges = $12a$

Here, a = length of the side

CUBOID

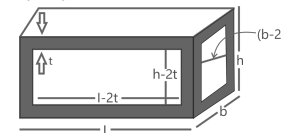
A rectangular body having 3D rectangular shape, is called a cuboid.



- Volume = $l \times b \times h$
- Total surface area = $2(lb + bh + lh)$
- Diagonal = $\sqrt{l^2 + b^2 + h^2}$

BOX

A box has its shape like a cube or cuboid. The amount that a box can hold or contain is called the capacity of the box. Capacity means internal volume.



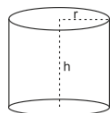
- Surface area of an open box = $2(\text{length} + \text{breadth}) \times \text{height} + \text{length} \times \text{breadth} = 2(l + b) \times h + l \times b$
- Capacity of box = $(l - 2t)(b - 2t)(h - 2t)$; where, t = thickness of box
- Volume of the material of the box = External volume - Internal volume (or capacity) = $lbh - (l - 2t)(b - 2t)(h - 2t)$

ROOM

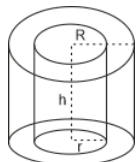
A rectangular room has four walls (surfaces) and opposite walls have equal

area.

- Total Area of walls = $2(l + b) \times h$
- Total volume of the room = $l \times b \times h$
- Area of floor or roof = $l \times b$

CYLINDER

- Volume of cylinder = area of base $\times$ height = $\pi r^2 h$
- Curved surface area = Perimeter of base $\times$ height = $2\pi r h$
- Total surface Area = curved surface area + area of both the circles = $2\pi r h + 2\pi r^2 = 2\pi(r + h)r$
- When the rectangular sheet is folded along its length, then the length becomes the circumference of the base of the cylinder and breadth becomes the height of the cylinder.

HOLLOW CYLINDER

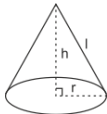
- Volume of hollow cylinder = $\pi(R^2 - r^2)h$
- Curved surface area = $2\pi(R + r)h$
- Total surface area = $2\pi(R + r)h + 2\pi(R^2 - r^2)$
 $= 2\pi(R + r)\{h + R - r\}$
 Where, R = External radius of cylinder, r = internal radius of cylinder, h = height

PRISM

- Volume of prism = area of base $\times$ height
- Lateral surface area = Perimeter of base $\times$ height
- Total surface Area = Lateral surface area + area of base and top surface

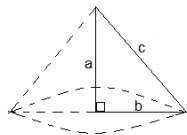
CONE

A solid and round body with a round base and pointed peak.



$$\text{Volume} = \frac{1}{3} \times \text{base area} \times \text{height} = \frac{1}{3} \pi r^2 h$$

- Slant height (l) = $\sqrt{r^2 + h^2}$
- Curved surface area = $\pi r l$
- Total surface area = $\pi r l + \pi r^2 = \pi r(l + r)$
- Cone formed by rotating right angled triangle about its height:



- Volume of cone so formed = $\frac{1}{3} \pi b^2 a$
- Similarly, Cone formed by rotating right angled triangle about its base:
- Volume of cone so formed = $\frac{1}{3} \pi a^2 b$
- Similarly, Cone formed by rotating right angled triangle about its hypotenuse:

$$\text{Volume of cone so formed} = \frac{1}{3} \pi r^2 c, \text{ (where } r \text{ is the altitude on hypotenuse and } r = \frac{a \times b}{c} \text{)}$$

Note: If the base is not round, it will be called a pyramid. A pyramid can have various shapes of the base example: square, rectangular, triangular etc.

PYRAMID

- Pyramid means a structure with regular polygon as its base and sloping sides that meet in a point at the top.
- In Pyramid, with n sided regular polygon at its base, total number of vertices = $n + 1$
- Volume = $= \frac{1}{3} \times \text{base area} \times \text{height}$

$$\begin{aligned} \text{Slant height (l)} &= \sqrt{r^2 + h^2} \\ \text{Lateral surface area} &= \frac{\text{Perimeter} \times \text{slant height}}{2} \end{aligned}$$

- Total Surface area = Lateral surface area + Area of base

TETRAHEDRON

It is a 3D figure made by joining four equilateral triangles.

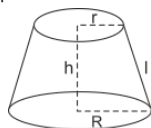


- Volume (V) = $\frac{1}{3} \times \text{base area} \times \text{height}$
 $= \frac{1}{3} \times \frac{\sqrt{3}}{4} a^2 \times \frac{\sqrt{6}}{2} a = \frac{\sqrt{2}}{12} a^3$
Remember this formula!

- Lateral surface area = $3 \times \frac{\sqrt{3}}{4} a^2$
- Total surface area = $4 \times \frac{\sqrt{3}}{4} a^2 = \sqrt{3} a^2$
- Height of Tetrahedron = $\frac{\sqrt{6}}{3} a$

FRUSTUM OF CONE

If a cone is cut by a plane parallel to its base, so as to divide the cone into two parts: upper part and lower part, then the lower part is called frustum.



- Slant height (l) = $\sqrt{h^2 + (R - r)^2}$
- Curved Surface Area = $\pi(R + r)l$
- Total surface area = $\pi(R + r)l + \pi R^2 + \pi r^2 = \pi\{(R + r)l + R^2 + r^2\}$
- Volume = $\frac{1}{3} \pi h(r^2 + R^2 + rR)$

SPHERE

- Volume of sphere = $\frac{4}{3} \pi r^3$
- Curved Surface area = Total surface area = $4\pi r^2$

HOLLOW SPHERE OR SPHERICAL SHELL:

- Volume of hollow sphere = $\frac{4}{3} \pi(R^3 - r^3)$
- Internal surface area = $4\pi r^2$
- External surface area = $4\pi R^2$
 Here R = external radius and r = internal radius

HEMISPHERE

- Volume of the hemisphere = $\frac{2}{3} \pi r^3$
- Total surface area = $3\pi r^2$
- Curved surface area = $2\pi r^2$
 Where, r = radius

IMP. UNIT CONVERSION :

- $1 \text{ m}^3 = 1000 \text{ litres}$ and
- $1 \text{ litre} = 1000 \text{ cm}^3$
- $1 \text{ meter} = 10 \text{ decimeter}$
 $= 100 \text{ cm} = 1000 \text{ millimeter}$
- $1 \text{ meter} = 10^{-1} \text{ decameter}$
 $= 10^{-2} \text{ hectometer}$
 $= 10^{-3} \text{ kilometer}$

Questions :-

Q.1. A circular pizza having a radius of 21 cm. If 80% of it is eaten, what is the area of pizza remaining?
 SSC CGL 12/09/2025 (Shift 1)
 (a) 240.42 cm² (b) 356.2 cm²
 (c) 277.2 cm² (d) 266.2 cm²

Q.2. A ring-shaped disc has outer radius 10 cm and inner radius 8 cm. What is the approximate ratio of the ring's area to the whole outer circle?
 SSC CGL 12/09/2025 (Shift 1)
 (a) 9 : 25 (b) 6 : 25 (c) 9 : 20 (d) 3 : 15

Q.3. A truck wheel having a radius of 42 cm. What percentage of its circumference is covered in a quarter turn? (Use $\pi = \frac{22}{7}$)
 SSC CGL 12/09/2025 (Shift 1)
 (a) 35% (b) 25% (c) 40% (d) 45%

Q.4. A circular park having a radius of 15 m. If a 1.5 m wide path is built around it, what is the approximate area of the path?
 SSC CGL 12/09/2025 (Shift 2)
 (a) 148.5 m² (b) 158.5 m²
 (c) 138.5 m² (d) 118.5 m²

Q.5. A right circular cone has a diameter of 14 cm and a height of 24 cm. What is the slant height?
 SSC CGL 12/09/2025 (Shift 3)
 (a) 25 cm (b) 30 cm (c) 35 cm (d) 40 cm

Q.6. A triangular prism has base area of 25 cm². If height is increased by 25%,
 SSC CGL 12/09/2025 (Shift 3)

Q.7. A circular disc's radius is decreased by 20%. What is the percentage decrease in its area?
 SSC CGL 12/09/2025 (Shift 3)
 (a) 20% (b) 38% (c) 36% (d) 30%

Q.8. A man has 160 m of fencing wire. He can use this wire to enclose a circular garden or a square garden. If he wants to maximize the enclosed area, what is the approximate ratio of the area of the largest possible circular garden to the area of the largest possible square garden, using the entire 160 m of wire?
 SSC CGL 12/09/2025 (Shift 3)
 (a) 1.3 : 1 (b) 1.27 : 1 (c) 1.7 : 1 (d) 1.6 : 1

Q.9. A vertical cylindrical container is filled with oil. A solid hemispherical stone of radius 6 cm is immersed completely, and the oil level rises by 2.5 cm. What is the radius of the cylinder?
 SSC CGL 12/09/2025 (Shift 3)
 (a) 6.29 cm. (b) 7.59 cm
 (c) 8.79 cm (d) 11.41 cm

Q.10. A sector of a circle has a central angle of 135° and a radius of 9 cm. Another sector of the same circle has a central angle of $\frac{3\pi}{4}$ radians. What is the ratio of the area of the first sector to the area of the second sector?
 SSC CGL 12/09/2025 (Shift 3)
 (a) 4 : 3 (b) 5 : 4 (c) 1 : 1 (d) 6 : 7

Q.11. The area of a sector of a circle with radius 21 cm and central angle 120° is:
 SSC CGL 13/09/2025 (Shift 1)
 (a) 693 cm² (b) 715 cm²
 (c) 645 cm² (d) 750 cm²

Q.12. What is the area of a sector with a central angle of 60° and radius 4 cm?
 SSC CGL 13/09/2025 (Shift 1)
 (a) $\frac{8\pi}{3}$ cm² (b) $\frac{5\pi}{3}$ cm²
 (c) $\frac{\pi}{4}$ cm² (d) $\frac{5\pi}{6}$ cm²

Q.13. A ring-shaped metallic sheet has outer and inner radii 14 cm and 7 cm. What percentage of the full circle's area is the ring?
 SSC CGL 13/09/2025 (Shift 2)
 (a) 36% (b) 48% (c) 64% (d) 75%

Q.14. A circular pond is surrounded by a 2 m wide path. If the total area (pond + path) is 804.25 m² and the area of the pond alone is 615.75 m², what is the radius (approximately) of the pond?
 SSC CGL 13/09/2025 (Shift 2)
 (a) 10 m (b) 14 m (c) 12 m (d) 11 m

Q.15. The ratio of the circumferences of two circular flower beds is 5 : 4. If the smaller flower bed has a radius of 10 m, what is the radius of the larger flower bed?
 SSC CGL 13/09/2025 (Shift 2)
 (a) 8 m (b) 7 m (c) 9 m (d) 5 m

Q.16. A cylinder and a cone have the same base radius and volume. Find the ratio of the height of the cylinder to the height of the cone.
 SSC CGL 13/09/2025 (Shift 3)
 (a) 1 : 3 (b) 3 : 1 (c) 2 : 1 (d) 1 : 2

Q.17. Two circular ponds have circumferences in the ratio 6 : 10. If the smaller pond has an area of 160 m², what is the approximate area of the larger pond?
 SSC CGL 13/09/2025 (Shift 3)
 (a) 444 m² (b) 450 m²
 (c) 438 m² (d) 449 m²

Q.18. If the base of a prism is a regular hexagon of side 3 cm and the height is 8 cm, what is its volume?
 SSC CGL 13/09/2025 (Shift 3)
 (a) $106\sqrt{3}$ cm³ (b) $104\sqrt{3}$ cm³
 (c) $107\sqrt{3}$ cm³ (d) $108\sqrt{3}$ cm³

Q.19. Three concentric circles are designed as part of a target practice board. Their diameters form an arithmetic progression. If the smallest circle has a circumference of 8π cm and the largest circle has a circumference of 16π cm, what is the circumference of the middle circle?
 SSC CGL 13/09/2025 (Shift 3)
 (a) 14π cm (b) 12π cm
 (c) 10π cm (d) 11π cm

Q.20. A sector has a central angle of 225° and a radius of 6 cm. Another sector of the same circle has a central angle of $\frac{5\pi}{4}$ radians. What is the ratio of the area of the first sector to the area of the second sector?
 SSC CGL 13/09/2025 (Shift 3)
 (a) 5 : 7 (b) 1 : 1 (c) 4 : 3 (d) 3 : 7

Q.21. A circular garden having a diameter of 20 m is surrounded by a concrete path that is 2 m wide. Calculate the percentage increase in area resulting from the addition of the path.
 SSC CGL 14/09/2025 (Shift 1)
 (a) 24% (b) 20% (c) 44% (d) 54%

what is the new volume (original height was 16 cm)?
 SSC CGL 12/09/2025 (Shift 3)
 (a) 550 cm³ (b) 575 cm³
 (c) 500 cm³ (d) 225 cm³

Q.7. A circular disc's radius is decreased by 20%. What is the percentage decrease in its area?
 SSC CGL 12/09/2025 (Shift 3)
 (a) 20% (b) 38% (c) 36% (d) 30%

Q.8. A man has 160 m of fencing wire. He can use this wire to enclose a circular garden or a square garden. If he wants to maximize the enclosed area, what is the approximate ratio of the area of the largest possible circular garden to the area of the largest possible square garden, using the entire 160 m of wire?
 SSC CGL 12/09/2025 (Shift 3)
 (a) 1.3 : 1 (b) 1.27 : 1 (c) 1.7 : 1 (d) 1.6 : 1

Q.9. A vertical cylindrical container is filled with oil. A solid hemispherical stone of radius 6 cm is immersed completely, and the oil level rises by 2.5 cm. What is the radius of the cylinder?
 SSC CGL 12/09/2025 (Shift 3)
 (a) 6.29 cm. (b) 7.59 cm
 (c) 8.79 cm (d) 11.41 cm

Q.10. A sector of a circle has a central angle of 135° and a radius of 9 cm. Another sector of the same circle has a central angle of $\frac{3\pi}{4}$ radians. What is the ratio of the area of the first sector to the area of the second sector?
 SSC CGL 12/09/2025 (Shift 3)
 (a) 4 : 3 (b) 5 : 4 (c) 1 : 1 (d) 6 : 7

Q.11. The area of a sector of a circle with radius 21 cm and central angle 120° is:
 SSC CGL 13/09/2025 (Shift 1)
 (a) 693 cm² (b) 715 cm²
 (c) 645 cm² (d) 750 cm²

Q.12. What is the area of a sector with a central angle of 60° and radius 4 cm?
 SSC CGL 13/09/2025 (Shift 1)
 (a) $\frac{8\pi}{3}$ cm² (b) $\frac{5\pi}{3}$ cm²
 (c) $\frac{\pi}{4}$ cm² (d) $\frac{5\pi}{6}$ cm²

Q.13. A ring-shaped metallic sheet has outer and inner radii 14 cm and 7 cm. What percentage of the full circle's area is the ring?
 SSC CGL 13/09/2025 (Shift 2)
 (a) 36% (b) 48% (c) 64% (d) 75%

Q.14. A circular pond is surrounded by a 2 m wide path. If the total area (pond + path) is 804.25 m² and the area of the pond alone is 615.75 m², what is the radius (approximately) of the pond?
 SSC CGL 13/09/2025 (Shift 2)
 (a) 10 m (b) 14 m (c) 12 m (d) 11 m

Q.15. The ratio of the circumferences of two circular flower beds is 5 : 4. If the smaller flower bed has a radius of 10 m, what is the radius of the larger flower bed?
 SSC CGL 13/09/2025 (Shift 2)
 (a) 8 m (b) 7 m (c) 9 m (d) 5 m

Q.16. A cylinder and a cone have the same base radius and volume. Find the ratio of the height of the cylinder to the height of the cone.
 SSC CGL 13/09/2025 (Shift 3)
 (a) 1 : 3 (b) 3 : 1 (c) 2 : 1 (d) 1 : 2

Q.17. Two circular ponds have circumferences in the ratio 6 : 10. If the smaller pond has an area of 160 m², what is the approximate area of the larger pond?
 SSC CGL 13/09/2025 (Shift 3)
 (a) 444 m² (b) 450 m²
 (c) 438 m² (d) 449 m²

Q.18. If the base of a prism is a regular hexagon of side 3 cm and the height is 8 cm, what is its volume?
 SSC CGL 13/09/2025 (Shift 3)
 (a) $106\sqrt{3}$ cm³ (b) $104\sqrt{3}$ cm³
 (c) $107\sqrt{3}$ cm³ (d) $108\sqrt{3}$ cm³

Q.19. Three concentric circles are designed as part of a target practice board. Their diameters form an arithmetic progression. If the smallest circle has a circumference of 8π cm and the largest circle has a circumference of 16π cm, what is the circumference of the middle circle?
 SSC CGL 13/09/2025 (Shift 3)
 (a) 14π cm (b) 12π cm
 (c) 10π cm (d) 11π cm

Q.20. A sector has a central angle of 225° and a radius of 6 cm. Another sector of the same circle has a central angle of $\frac{5\pi}{4}$ radians. What is the ratio of the area of the first sector to the area of the second sector?
 SSC CGL 13/09/2025 (Shift 3)
 (a) 5 : 7 (b) 1 : 1 (c) 4 : 3 (d) 3 : 7

Q.21. A circular garden having a diameter of 20 m is surrounded by a concrete path that is 2 m wide. Calculate the percentage increase in area resulting from the addition of the path.
 SSC CGL 14/09/2025 (Shift 1)
 (a) 24% (b) 20% (c) 44% (d) 54%